

Understanding Employee Concerns and Misconceptions About AI in SMEs

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Abstract

As artificial intelligence (AI) continues to shock the world with its limitless capabilities, many companies have been taking advantage of this new technology, using it for the purpose of improving their company. And as it continues to integrate into various business operations, it is crucial to understand employee perceptions of this technology, specifically what kind of concerns and misconceptions these employees hold about AI that might be holding back their acceptance of AI in their personal and professional lives. By discussing fears such as job replacement, data privacy issues, and the impact of AI on creativity, through qualitative data collection made with interviews done to employees from small to medium-sized enterprises (SMEs), the study reveals that the portrayal of AI in the media and a lack of understanding contribute to employee hesitation regarding the adoption of AI. The results suggest a need for better education about AI within the company, increased transparency, and the use of AI tools in a way that complements human skills rather than replacing them. By solving these issues, SMEs can foster a greater acceptance of AI technologies within SMEs, ending in an effective integration and use of AI.

Keywords: Artificial Intelligence, concerns, SMEs (Small to medium-sized enterprises), lack of understanding, job replacement, data privacy

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1. Introduction

Artificial intelligence (AI) has gone from a niche technology only specialists would know to a worldwide known topic that has captured the attention of industry professionals as well as the public. The rapid advancements of AI have far exceeded all expectations by demonstrating its capabilities to reshape industries. Now that AI has seemingly integrated into the daily lives of people, particularly in professional workplaces, it is important to understand how employees perceive this shift and what they think of this new innovative technology. While many notice the benefits of AI, the emergence of potential concerns and misconceptions is a worrying fact that needs to be addressed.

What this study explores are the concerns and misconceptions employees in small to medium-sized enterprises (SME's) have about AI. Concerns such as job replacement and automation, manipulation from the media, or even fictional portrayals of AI in pop culture. These kinds of fears, real or fictional, can be a breeding ground for uncertainty and hesitation in the adoption of AI technologies.

This study aims to investigate the concerns and misconceptions that employees in SMEs have about AI, or lack thereof. Specifically, this research seeks to answer the question: **"What are the concerns or misconceptions employees from SMEs have about AI?"** By collecting qualitative data from willing participants, this research will explore the driving factors of these perceptions and aim to find the reasoning behind some of the employees' apprehensiveness.

1.1. Research Hypotheses

Based on the initial exploration of employee concerns and misconceptions about AI, this study proposes the following hypotheses:

- Hypothesis 1:** Employees will be more willing to adopt AI into their workplace if it will benefit them and improve their job performance.
- Hypothesis 2:** Employees with less affinity to technology and confidence in applying AI concepts will see the complexity of AI as a significant barrier to learning.
- Hypothesis 3:** Employees in SMEs are less likely to support the adoption of AI if they perceive that AI technologies pose a threat to job security.

- Hypothesis 4:** Negative portrayals of AI in media and pop culture significantly contribute to the reluctance of employees in SMEs to engage with AI technologies.

2. Literary review

Oldemeyer, Jede, & Teuteberg (2024) [3], has recently done research into the challenges of adopting AI into the SME space. Their paper has highlighted several of them, including knowledge gaps, high costs, and inadequate infrastructure. Many SMEs lack the skilled employees or an understanding of AI and how to effectively apply it to their companies. A big misconception is the fact that AI can only be used by large companies, along with concerns over return on investment (ROI), all of which contribute to the hesitation that SMEs have when adopting this new technology. Adding to that, most SMEs have limited financial resources, making it even harder. Technologically speaking, most SMEs have low digital maturity, missing the necessary infrastructure and data capabilities needed to successfully integrate AI.

The paper also points out another barrier to implementation, which is managerial hesitation. Some SME leaders/owners view AI as unnecessary or too complex for their operations. These factors lead to resistance to the adoption of AI, even if they recognize the potential benefits in operations and efficiency. Going beyond business operations, other sectors such as HR have yet to be touched by AI, signaling a need for more research and more practical, SME-focused case studies. [3]

Szedlak, Poetters, & Leyendecker (2020) [1] believe that some SMEs are concerned about the possible negative influence AI might have over the company, fearing that if AI is improperly implemented into business operations, it could lead to loss of value rather than value generation. They might think this is due to misaligned AI applications, meaning they might not be suited to the businesses' regular operations and model, which could lead to inefficiencies or damage done to the competitive edge. Other reasons include the previously mentioned high costs, low ROI, and skill gaps. Those reasons delay the adoption of AI in SMEs, adding to the list of concerns employees have regarding AI.

Kallmuenzer, Mikhaylov, Chelaru, and Czakon (2024) [2] suggest that technological path dependency is a concern for SMEs. These

companies have relied on the same technologies for years and, at this point in time, are unwilling to transition into a newer technology such as AI. They underestimate the enormous benefits that AI can bring to them and overestimate the costs needed to implement it. And much like all the other research that was looked at, this paper also implies that a lack of digital skills within SMEs is a big factor and concern for them, as well as financial limitations, which play a big part. Many of these companies all have the same issues and limitations stopping them from implementing AI, which overall might make the employees working at these SMEs less open-minded about AI since their company won't implement it yet, and them having to face the same hurdles to get into using AI in their daily lives discourages them.

All these concerns that SMEs raise stem from risk aversion. They want the best for their company and employees, and for many of them, implementing something so new and something that they have no understanding of is not worth it, not mentioning all the other limitations like budgetary issues. This paper focuses more on the individual opinions and concerns of employees, but understanding why the SMEs and the management themselves don't want the implementation of AI is key to understanding the reasoning behind the employee's' hesitation to adoption.

3. Methodology

3.1. Qualitative methods

All of this paper is based off the qualitative data that was collected throughout this project, since the research question is in nature qualitative, making this dependent on qualitative analysis and the data extracted from that.

3.1.1. Qualitative data collection

For the qualitative data collection methods, it was all done through semi-structured interviews conducted by the whole research team, one participant per team member. Before conducting any type of interview, an interview plan was made, which included methods of analyzing qualitative data and the set questions written for the participants to answer. The questions totaled 13, making the average duration of each interview be about 20 to 25 minutes. The questions were designed to get the most data and information out of the participants regarding the goal of the study, which was to get the highest quality type of data for the sake of the study. The interviews also included a mix of in-person interviews and online interviews, both being acceptable forms of interviewing and serving the same purpose. Each interview was recorded so that the information could be accessed later. For the online interviews, it was using the built-in record function that most video meeting platforms have, which is later exported to MP3 to be used. For the in-person interviews, a voice recorder on the phone was used to record and then exported to MP3.

The total sample size of the qualitative data collection was 6, so there were 6 participants that were interviewed. Before each interview that was conducted, each participant was given an information letter that briefly informed them about what the study was about, what was going to be done with their data, and any other important information that participants had the right to know before taking part. After the letter, each participant was given another document called the letter of informed consent. This document allows participants to give their informed consent by ticking a few boxes and ending it with a signature from the participant and the lead researcher. After conducting the interviews and getting the consent from the participants to use their data in our studies, it's time to put this data to good use.

3.1.2. Qualitative analysis

After having conducted each interview, every audio recording was transcribed to text. Each participant was separated into speakers so

that it's clearer to read what each person said, easing the analyzation process. After transcribing everything one-by-one, each transcription was reviewed and assigned specific codes to different excerpts that fit the criteria of the code. So, if an excerpt was talking about fears of AI automation, it would be marked with a code named "automation". In total, across all transcripts, there were around 15 to 20 different codes that were gathered to mark the data. After doing that for each of them, each code would be grouped into more general themes, of which there were four. Grouping them into themes gives a more general look at the structure of the results of our research. See Figure 1 on page 3, the thematic analysis illustrates the key themes and their associated codes. Now that the data has been analyzed, a narrative is ready to be written to showcase the results.

3.1.3. Building a narrative

The next step after coding the data is to build a narrative based on the data, a way to show the results of the research and put the words onto paper that can be read and understood. All the themes of each transcript were reviewed and divided into its respective sections. Now comes the part about writing the narrative of each section individually, going through all the themes and excerpts that were collected. The interpretations of participants' responses were made whilst considering the research question, ensuring that the narrative was aligned with the broader context of the study, while strengthening the narrative by including direct quotes from the participants. With this, an idea of what the results will look like can be obtained.

3.2. Data Storage and Privacy

The data that was collected from the willing participants is securely stored in systems approved by the University located within the EU for a duration of 5 years, avoiding any non-compliant tools such as Google Drive, and the data is only accessible to members of the team and University staff. All data is anonymized and replaced with codes; no personal information will be used for any purpose. In cases where the participant wants to withdraw from the study, the data of the participant would be deleted and not used in the research, and there would be no consequence or explanation needed.

4. Results

4.1. Qualitative results

The research question delves into the concerns and misconceptions that employees of SMEs have about AI. And with that, having conducted all the interviews, a number of different concerns that employees and business owners alike have with them were found. The next section of the paper will explore the concerns they have without interpretation.

4.1.1. Lack of a human element

The first concern that came up in almost every interview was the lack of a human element. This concern refers to how AI lacks human emotions and can not perform any tasks that would require human intelligence, such as human judgment and empathy. One of the participants shared a story, saying he used ChatGPT to generate text for his customers, to which his boss said that he could not use that piece of generated text as they do not talk like that to their customers, and it would seem disingenuous. "Did you, did you really type this? I was like, no, of course not. So, he was just like, yeah, but we don't talk this way to our customers." Some of the participants also expressed doubts about AI being able to perform tasks that require empathy.

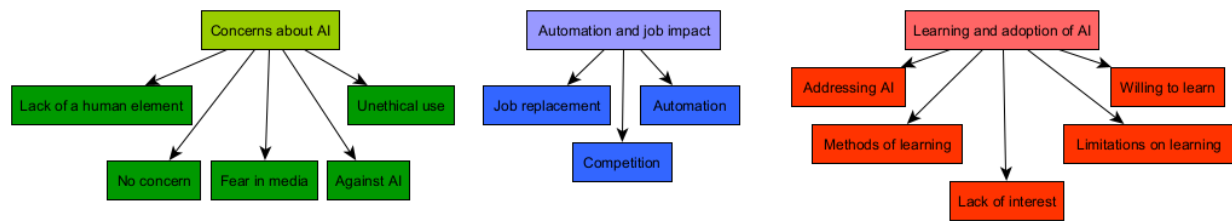


Figure 1. Thematic Analysis Diagram

4.1.2. AI misuse and Data Privacy concerns

Several participants expressed concerns regarding their own data and information when using ChatGPT and other forms of generative AI. Whenever they used ChatGPT, they wondered where it was being stored and for what it was being used. One of them asked, "If I put things in AI, like ChatGPT, where is it? Who has it?" Tying into that, they were also concerned about people using AI tools to do harm, specifically with other people's information and likeness. For example, using someone else's face and voice to create false AI videos, portraying them in an undesirable way.

4.1.3. Portrayal of AI in media and pop-culture

All participants acknowledged that the media significantly influences their perception of AI, often times leading them to have misconceptions about AI that were given to them by way of the media and pop-culture material. They realize the dangers of this, noting that information is too easily believed without question. One of the participants specifically mentioned that there is a strong case of that in his home country, the Netherlands. He went on to say, "If I see something on TV, like, uh, the typical Dutch, uh, average person... they believe it straight away. So, you see like, uh, it's dangerous,". He says that people here will believe most things that they see in the news or media. That is not the case for everything, but with statements that are more easily believable, they will believe in them.

4.1.4. Job Replacement

All participants expressed little to no concern about AI replacing their own jobs and positions at present. However, they all acknowledged that AI is already replacing many other roles and will likely continue to do so as companies increasingly automate tasks and positions using AI. Despite all of this, they do not perceive or feel any immediate danger of their current roles being replaced by AI. To add to it, several participants expressed that they would be more willing to learn about AI if it provided clear benefits to their work, such as improving efficiency or automating mundane tasks.

4.1.5. Problems with Generative AI

There were a few cases in which the participants expressed concerns that AI would become the main source of information for people. After having had bad experiences using Generative AI, giving them incorrect information or having misunderstood prompts, they concluded that it has still not reached full maturity and still requires significant improvements. One of them went on to state this, "If we start to see artificial intelligence as our source of information, so our main source of information, I think it's not transparent enough of how the algorithm and how the thoughts... are being put in there,". Another participant shared this with us, "People think that what they come up with themselves is not smart enough". Bringing in the concern that Gen AI makes us less creative as human beings because of our constant reliance on Gen AI.

5. Discussion

5.1. Qualitative discussion

This next section explores the same concerns mentioned in the results section, but this time it includes an interpretation of what each concern means, drawing conclusions for each concern.

5.1.1. Lack of a human element

Many of the participants expressed concerns over the lack of a human element. The data shows that the absence of human presence in AI bothers them. When they think about the fact that there is no human behind the responses that they receive from Chatbots is concerning to them, and they do not know how to feel about it. Talking about the example from before, when conversing with other humans via texts generated not from themselves but from an AI software, is something they do not feel comfortable doing as it is not them doing it or any other human but in fact a lifeless AI. This concern highlights a discomfort when relying on AI tools for communication, especially when elements like empathy and nuanced understanding are essential in many interactions. For many of them, the artificial nature of AI responses creates a barrier between people and AI, raising doubts over authenticity and emotional intelligence in AI systems.

5.1.2. AI misuse and Data Privacy concerns

The concerns and examples the participants provided highlight how dangerous AI can be if it ends up in the wrong hands. First, they showed concerns over where their data was being stored. Since they have little understanding of how this new technology works, they wondered where it is stored and what they were using their data for. This shows that the participants show little trust in AI systems, meaning that transparency on where personal data is being stored is important to build trust among users. But the bigger concern is how AI can be used to cause harm to others or themselves. The mentioned examples included deep-fakes, the use of replacing someone else's face, likeness and identity with something else, in a different situation showing them in a negative light. These concerns reflect a broader fear and ethical concern of how AI could be exploited to compromise privacy and security.

5.1.3. Portrayal of AI in media and pop-culture

The participants recognize that their views on AI are shaped by the media, which often emphasizes the negative aspects that AI has and creates unnecessary misconceptions. They understand that this constant exposure to the negative aspects and portrayals can lead to a never-ending cycle of mistrust and rejection toward AI. This suggests that the media has a massive influence over what perceptions people will have of AI and is ultimately contributing to the hesitation and fear surrounding AI and its adoption. This means if there is a stigma against AI in the media, the public watching will also hold that same opinion on it.

5.1.4. Job Replacement

The participants are currently not worried about their job security but are aware of the broader trend of automation across industries. As was mentioned before, they realize that AI is already replacing other job roles and automating industries and will continue to do so as businesses seek to automate wherever possible, taking advantage of this new technology. While they recognize this large shift, they don't feel like it poses any threat to their own positions, suggesting that they feel secure in their current roles, and they believe that AI still has a way to go before it's able to take over their current working positions and leave them out of a job. However, they know that it's starting to happen now. Employees also stated if AI would benefit them and improve work efficiency, they would be very willing to learn about AI. This indicates that employee motivation to learn AI hinges on seeing tangible improvements in their daily workflow.

5.1.5. Problems with Generative AI

Participants expressed deeper concerns about AI becoming the primary source of information, revealing their unease about the technology's current limitations and its lack of transparency. Because of their bad experiences using AI and Gen AI, they deemed it not to be fully mature yet. This suggests a broader mistrust over the technology's accuracy and transparency, leading to their doubts over the information it provides and their unease that people would start using that information on a daily basis. Another fear was that over-reliance on AI could stifle human creativity, leading people to undervalue their own intelligence. This suggests that, despite the convenience it provides, it might reduce the independent thinking of people and human innovation over time.

5.2. Addressing Hypotheses

5.2.1. Addressing Hypothesis 1

This hypothesis is supported by the research and results of the study. Many of the participants showed more of an interest in adopting AI into their workspace if it provided beneficial outcomes to them, provided that the tool complemented rather than replace human skills.

5.2.2. Addressing Hypothesis 2

This hypothesis is partially supported by the research. Employees who had limited knowledge of AI and less technologically adept expressed concerns regarding the complexity of AI and the adoption of it, viewing it as a challenge. However, a few participants stated that with adequate training and a little bit of hands-on experience, they would be willing to use AI tools, meaning that this hesitation can be mitigated.

5.2.3. Addressing Hypothesis 3

This hypothesis was not fully supported. While participants were aware that AI is capable of replacing certain roles, they did not feel an immediate threat to their current jobs. Instead, they expressed a sense of security in their existing roles and believed that AI was not yet advanced enough to replace them. They did acknowledge the broader industry trends towards automation but did not feel that their personal job security was at risk, suggesting that job replacement concerns did not strongly deter them from supporting AI adoption.

5.2.4. Addressing Hypothesis 4

The fourth and final hypothesis is fully supported by the data. Every participant frequently mentioned and admitted to how the media does in fact have an influence on how they view AI. Many participants indicated that the media tends to overemphasize the dangers of AI, contributing to their hesitation and mistrust of AI technologies.

5.3. Comparison with existing literature

The findings of this study align with the existing literature on AI concerns within SMEs. The paper made by Oldemeyer et al. (2024) [3] highlighted issues such as knowledge gaps, high costs and inadequate infrastructure, which were confirmed by showing the concerns of employees about their lack of skills and understanding of AI, which is needed to successfully implement AI. In addition, the hesitation shown by the management and employees that was picked up on in the literature review is reflected in how the participants felt about things like job replacement, data privacy and the lack of a human element, which are all concerns that came up in this study.

One of the points made by past research is how SME owners often find the adoption of AI within their company to be too complex and unnecessary. [3] This aligned with what some of the participants have said, one of them who is an SME owner stated this, "There's AI, it can help me. But how, how can it help me? What, what can it do?" and "Where is the knowledge? Where can I get the knowledge, that's the main problem". And another participant stated they didn't feel the need to implement AI to their business operations, as none of their employees have any interest in the technology, making AI an unnecessary element to his company.

Szedlak, Poetters, and Leyendecker (2020) [1], much like in this study, called attention to a fear that SMEs share the concern that improperly implemented AI could result in damage being done to the company's competitive edge and bring inefficiencies to the business's regular operations. This study supports this concern, as participants showed doubts and worries over AI's ability to perform tasks that require empathy and human judgment. And the point where participants had reservations about AI-generated content, such as generated texts not sounding empathetic and ingenuine, falls in line with the risks of misusing AI technology for malicious purposes that show up in the previous research.

Overall, this study confirms the well documented research done on this topic while providing real insights from individual employees belonging to SMEs. Their perspectives highlighted that the challenges that AI adoption into SMEs face, goes beyond financial and technical barriers, as psychological and cultural factors such as media influence, fear of job replacement, fear of AI capability and concerns over data privacy all play a role in shaping AI adoption attitudes within SMEs, which fills a gap present in other literature which always focused on financial and technical challenges without factoring in the employees themselves.

5.4. Limitations

While this study provides valuable insights into the concerns and misconceptions that employees from SMEs have about AI, there were a number of limitations that must be acknowledged. First, the sample size that was worked with was of only 6 participants, which limits the validity of the results and data. Although qualitative data and analysis emphasize more quality in data while having significantly fewer data points, having a larger and more diverse sample might have revealed an additional number of concerns that could have been used for the sake of the study.

Another limitation is interviewer bias, where wording and tone of questions might have influenced the participants' responses when conducting the semi structured interviews. During the interviews, we the interviewers tried to make the questions fair with no interpretation or personal skewing involved, with the delivery of the questions being relaxed and not trying to force any kind of biased response out of the participants. Even with all of that, biases might still have made its way into the results.

A lack of time is a limitation that had to be acknowledged, as we only had give or take 8 weeks to complete the entire project. If there

was more time to work with, then the project might have had more insights to give, and the paper might have been more refined.

Lastly, this study does not account for the rapidly changing environment that AI lives in. As AI continues to advance, the concerns and misconceptions may very well change, meaning further research would be needed to maintain pace with these technological advancements.

6. Conclusion

In conclusion, this study outlines several common concerns and misconceptions that employees of SMEs have about AI. What the findings show is that although these employees are aware of the potential that AI holds, they are also cautious, particularly regarding issues like job replacement, data privacy, and the lack of a human element in AI interactions. Additionally, the portrayal of AI in the general media and pop-culture material has a huge impact on how AI is viewed in the minds of not only employees of SMEs but also the general public, which all fuels the uncertainty and hesitation in adopting AI technologies.

The results confirm that the efforts to address AI adoption should not only be directed towards the practical barriers and cost of implementation of AI but also towards dispelling the concerns and misconceptions of SME employees and building trust around AI by educating the employees and using transparent AI systems.

Tackling these concerns requires, as it was mentioned, increasing transparency in AI systems, improving education on AI's capabilities and downsides, and promoting AI in such a way that it benefits and complements human skills rather than completely replacing them. This way, SMEs can help their employees understand more about AI, start to be more open to using it and enjoying the benefits that AI has to offer. This study provides the base for understanding negative AI perception among employees and includes valuable insights for introducing a new-found trust and acceptance in this technology.

6.1. Acknowledgments

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7. Contact

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